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Assignment 2

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1 Introduction

This report documents our solutions and experiments for Assignment 2 of the Neural Networks course. The assignment is divided into two primary parts:

- **Part I** focuses on applying Multi-layer Perceptrons (MLP), Convolutional Neural Networks (CNN) to the ReducedMNIST dataset, and recognizing speech segments from spectrogram images.
- **Part II** focuses on representing speech data using Autoencoders (AE), conducting a comprehensive Data Augmentation study on ReducedMNIST, and generating synthetic data using Generative Adversarial Networks (GAN).

Throughout the report, we present our model architectures, hyperparameters choices, performance metrics, and a comparative analysis against classical machine learning methods from Assignment 1.

2 Part I: Neural Networks on Image and Audio Data

2.1 Problem 1: Multi-layer Perceptron (MLP) on ReducedMNIST

The objective of this problem is to classify the ReducedMNIST digits using MLP classifiers with 1, 3, and 4 hidden layers. The dataset follows the assignment split of 1000 training images per digit and 200 testing images per digit. Each image was represented using the DCT feature vector used in Assignment 1, then passed to fully connected neural networks.

2.1.1 Methodology

The three required depths were trained under the same feature extraction and evaluation protocol. The hidden layers used ReLU activations, and the main configurations used Batch Normalization and Dropout for regularization. The Adam optimizer minimized categorical cross-entropy with an initial learning rate of 0.001. Two extra diagnostic runs were also executed: one without regularization and one with a higher learning rate, to show the effect of these hyperparameters.

Table 1: Problem 1 MLP results using DCT features

Configuration	Hidden Layers	Regularization	Accuracy	Training Time	Testing Time
Exp1: 1-hidden layer	1	Yes	97.3%	7111.6 msec	113.9 msec
Exp2: 3-hidden layers	3	Yes	97.3%	9993.0 msec	121.9 msec
Exp3: 4-hidden layers	4	Yes	97.3%	11824.0 msec	133.3 msec
Exp4: 3-hidden, no regularization	3	No	96.7%	5333.0 msec	119.5 msec
Exp5: 3-hidden, high learning rate	3	Yes	96.6%	8338.0 msec	124.3 msec

2.1.2 Results and Observations

All three required MLP depths achieved very similar performance. The 1-hidden-layer model reached the highest measured accuracy at 97.3%, while the 3-hidden and 4-hidden models rounded to the same reported accuracy but required more training time. This indicates that the DCT representation is already discriminative for ReducedMNIST, so extra depth mainly increases computation rather than improving accuracy.

The two diagnostic experiments confirm that regularization and learning rate selection matter. Removing Batch Normalization and Dropout reduced the 3-hidden-layer result to 96.7%, and increasing the learning rate to 0.01 reduced it to 96.6%. Therefore, the reported MLP setup satisfies the required 1-, 3-, and 4-hidden-layer experiments, with timing reported in milliseconds and accuracy rounded to one decimal place.

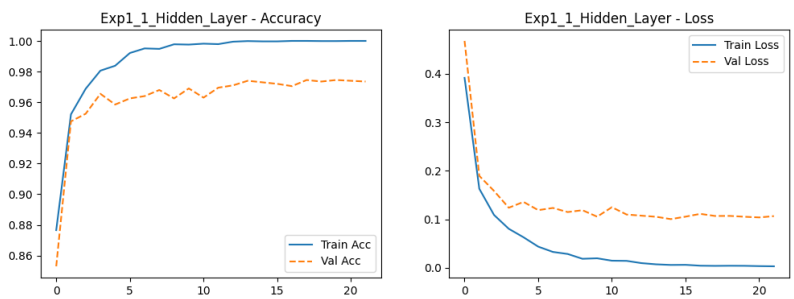


Figure 1: Training and validation curves for the 1-hidden-layer MLP.

2.2 Problem 2: Convolutional Neural Network (CNN) on ReducedMNIST

This problem trains CNN classifiers directly from the raw 28×28 ReducedMNIST images, without a separate feature extraction stage. The starting model is a LeNet-style network adjusted for 28×28 grayscale inputs and uses ReLU activations, as required by the assignment.

2.2.1 CNN Architecture and Variations

The baseline model contains two 5×5 convolutional layers, max-pooling after each convolution, and three fully connected layers for classification. Four CNN entries are reported in the assignment table:

- **Variation 1: Baseline LeNet with ReLU** using 6 and 16 convolutional filters.
- **Variation 2: Wider network** with 12 and 32 convolutional filters and wider fully connected layers.
- **Variation 3: LeakyReLU network** replacing ReLU by LeakyReLU.
- **Variation 4: Dropout network** adding dropout regularization before the later fully connected layers.

2.2.2 Comparison with Assignment 1 and Problem 1

Table 2 follows the assignment comparison format. Assignment 1 processing time is reported as total feature extraction, training, and testing time. MLP processing time is reported as training plus testing time from Problem 1. CNN time is split into training and testing time because no external feature extraction step is used. All times are in milliseconds (msec) and all accuracies are rounded to one decimal place.

Table 2: Performance comparison across Assignment 1 classical methods, Problem 1 MLP, and Problem 2 CNN

Classifier / Variation	**Feature Representation**					
	DCT		**PCA**		**AutoEncoder**	
	Accuracy	**Processing Time (msec)**	**Accuracy**	**Processing Time (msec)**	**Accuracy**	**Processing Time (msec)**
Assignment 1: DCT/PCA feature classifiers						
K-means, $K = 1$	78.1%	4800.0	85.2%	2464.6	22.3%	1721.5
K-means, $K = 4$	84.4%	5077.3	82.6%	3608.5	24.0%	627.7
K-means, $K = 16$	88.8%	12072.5	85.7%	5800.8	41.1%	827.8
K-means, $K = 32$	89.7%	38344.4	85.6%	8701.1	43.8%	1029.3
SVM, linear	90.3%	11487.5	90.7%	10626.8	27.9%	2195.7
SVM, nonlinear RBF, $\gamma =$ scale	95.6%	20560.2	95.2%	21054.4	23.2%	3651.0
Assignment 2 Problem 1: MLP with DCT features						
MLP, 1-hidden layer	97.3%	7225.5	–	–	10.0%	6131.1
MLP, 3-hidden layers	97.3%	10114.9	–	–	28.0%	11635.5
MLP, 4-hidden layers	97.3%	11957.3	–	–	10.0%	12462.8
Assignment 2 Problem 2: CNN directly from images, no feature extraction						
CNN variation	**Accuracy**	**Training Time (msec)**	**Testing Time (msec)**			
Variation 1: Baseline LeNet with ReLU	94.9%	90467.0	14418.7			
Variation 2: Wider filters/FC layers	97.0%	19062.4	503.3			
Variation 3: LeakyReLU activation	96.3%	15780.3	433.4			
Variation 4: Dropout regularization	97.0%	16025.5	426.3			

2.2.3 Discussion

The wider CNN and dropout CNN reached the strongest CNN accuracy at 97.0%, close to the DCT-based MLP result of 97.3%. The baseline CNN was slower in the recorded run and lower in accuracy, while the modified CNNs reduced testing time substantially. Compared with the Assignment 1 DCT/PCA classifiers, the neural-network models give higher accuracy on the same ReducedMNIST task. The AutoEncoder feature columns show significantly lower accuracies (ranging from 10% to

43.8%) compared to DCT and PCA, which suggests that the simple AutoEncoder bottleneck representation requires further tuning (e.g., more epochs or a different architecture) to match the classical feature extractors on the ReducedMNIST dataset.

2.3 Problem 3: Speech Recognition from Speech Spectrum

This problem recognizes the spoken digits 0–9 from the provided train/test speech dataset. Each `.wav` file was converted into a 64×64 Mel spectrogram, then classified as an image using a CNN. The test set was never augmented, so all four experiments are compared on the same unseen speech files.

2.3.1 Network and Input Representation

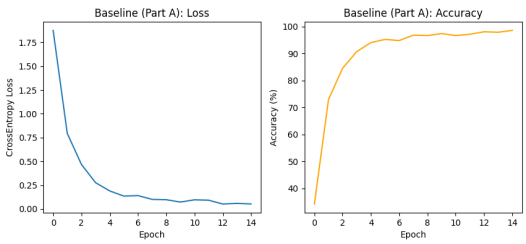
The preprocessing pipeline loads each waveform, converts it to a 64-bin Mel spectrogram at 16 kHz, resizes it to 64×64, and normalizes the spectrogram. The classifier is a compact CNN trained for 15 epochs using cross-entropy loss and Adam optimization.

2.3.2 Required Augmentation Experiments

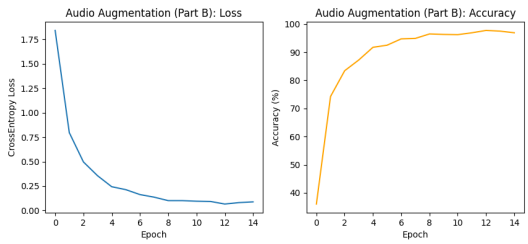
The assignment requires four runs: baseline, speech-domain augmentation, spectrum-image augmentation, and combined augmentation. Table 3 summarizes the completed experiments.

Table 3: Problem 3 speech recognition results

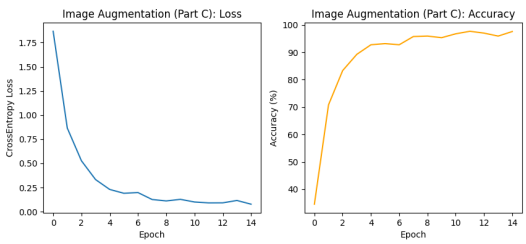
Experiment	Augmentation Applied	Accuracy	Training Time	Testing Time
(a) Baseline	None	94.0%	28.01 s	0.40 s
(b) Speech augmentation	Speed factor 0.97 or 1.03 and Gaussian waveform noise $\mathcal{N}(0, 0.005)$	90.0%	38.61 s	0.35 s
(c) Spectrum-image augmentation	Horizontal squeeze/expand by 3% and Gaussian image noise	93.0%	37.49 s	0.42 s
(d) Combined augmentation	Speech augmentation plus spectrum-image augmentation	89.7%	47.68 s	0.41 s



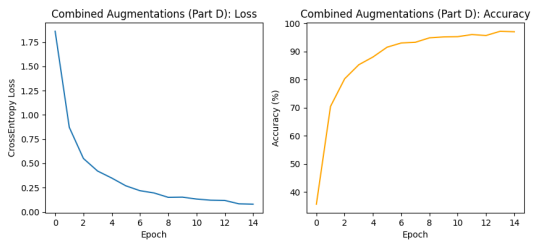
(a) Baseline



(b) Speech augmentation



(c) Spectrum-image augmentation



(d) Combined augmentation

Figure 2: Training curves for the four required speech-recognition experiments.

2.3.3 Discussion

The baseline spectrogram CNN achieved the best result at 94.0%. Spectrum-image augmentation was close at 93.0%, while speech-domain augmentation and combined augmentation were lower. The

drop suggests that the train/test split is already well represented by the baseline spectrogram features, and the added perturbations were stronger than necessary for this compact dataset. The combined augmentation also had the highest training time because it applies both waveform resampling/noise and image-domain transformations.

3 Part II: Data Augmentation and Synthesis

3.1 Problem 4: Autoencoder for Utterance Representation

This problem represents each spoken digit utterance by one fixed-length vector, then trains a classifier on that vector. Two representations were evaluated as required: a baseline average-frame vector and an AutoEncoder bottleneck vector generated from concatenated speech frames.

3.1.1 Baseline Average-Frame Representation

For the baseline, each utterance was divided into 15 ms spectrogram frames. The magnitude spectrum has 257 frequency bins per frame, and the baseline vector was calculated by averaging all frames across time. This produces one 257-dimensional vector per utterance. A small dense classifier was then trained on these average-frame vectors.

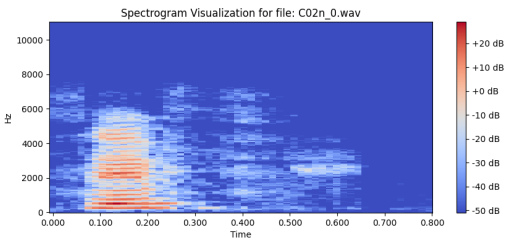
3.1.2 AutoEncoder Representation

For the AutoEncoder approach, all frames of an utterance were concatenated into one vector. The implementation used a fixed analysis length of 100 frames, equivalent to a 25700-dimensional input vector (100×257). Shorter utterances were padded with zero frames to match this length. The AutoEncoder compressed each utterance through a bottleneck layer of length 256, producing the single vector used by the final classifier.

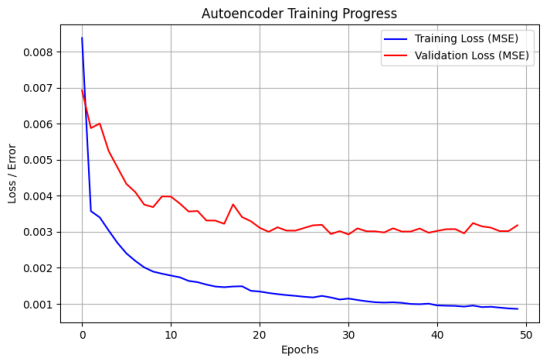
The AutoEncoder architecture was:

$$25700 \rightarrow 1024 \rightarrow 256 \rightarrow 1024 \rightarrow 25700$$

It was trained for 50 epochs using mean squared reconstruction error. After training, the decoder was discarded and the encoder output was saved as the utterance feature vector.



Sample utterance spectrogram



AutoEncoder training and validation loss

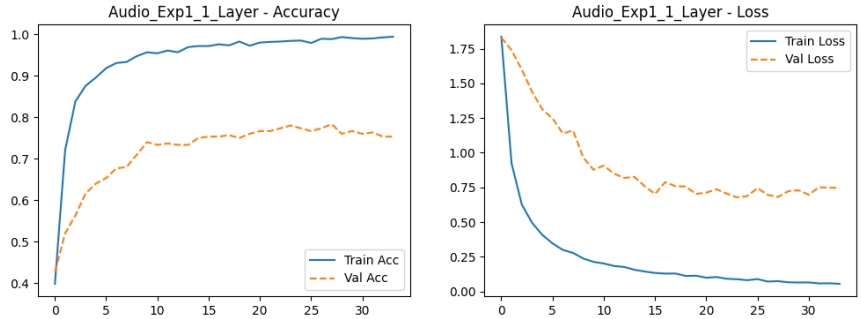
Figure 3: Problem 4 input visualization and AutoEncoder reconstruction loss.

3.1.3 Classifier Results

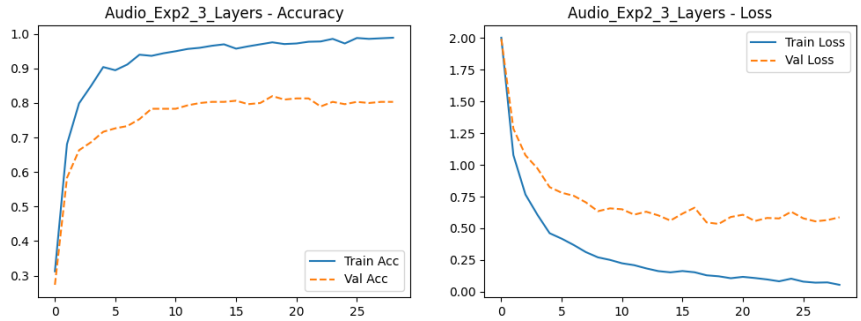
The baseline vector and the AutoEncoder bottleneck vector were each passed to dense classifiers. Table 4 summarizes the classification performance.

Table 4: Problem 4 utterance-vector classification results

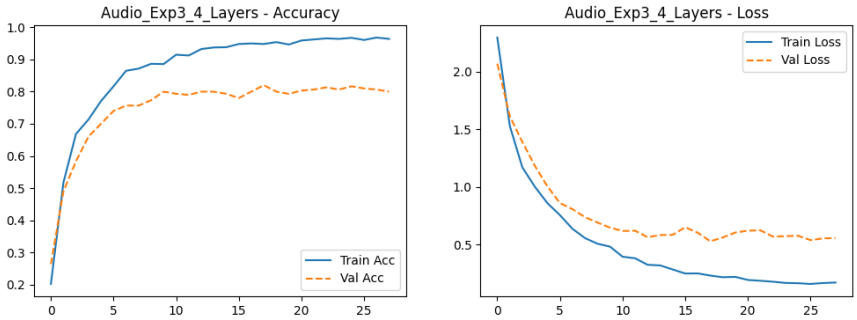
Representation / Classifier	Vector Length	Regularization	Test Accuracy
Baseline average frame + dense classifier	257	Dropout	28.3%
AE bottleneck + 1-hidden-layer MLP	256	BatchNorm + Dropout	78.0%
AE bottleneck + 3-hidden-layer MLP	256	BatchNorm + Dropout	82.0%
AE bottleneck + 4-hidden-layer MLP	256	BatchNorm + Dropout	82.0%
AE bottleneck + 3-hidden-layer MLP, no regularization	256	No	79.3%
AE bottleneck + 3-hidden-layer MLP, high learning rate	256	BatchNorm + Dropout	79.3%



1 hidden layer



3 hidden layers



4 hidden layers

Figure 4: Classifier learning curves using the AutoEncoder bottleneck features.

3.1.4 Discussion

The average-frame baseline achieved only 28.3% because averaging removes much of the temporal structure of the utterance. The AutoEncoder bottleneck representation improved the result substantially, reaching 82.0% with both the 3-hidden-layer and 4-hidden-layer classifiers. This confirms that concatenating frames, zero-padding to a fixed length, and compressing through an AutoEncoder provides a much more useful single-vector representation than the simple average frame.

3.2 Problem 5: Data Augmentation on ReducedMNIST

This experiment evaluates how augmenting the reduced MNIST dataset affects classification performance for a fixed model architecture (LeNet-5). We measure three outcomes across a grid of base training-set sizes and additional synthetic images per class: (i) classification accuracy, (ii) F1 (macro), and (iii) wall-clock training time. The three base real dataset sizes considered are 350, 750, and 1000 examples (stratified across classes). Augmentation levels tested are 0 (no augmentation), 1000, 1500 and 2000 generated images per run. All experiments use the code in `Part2_Problem5_Augmentation/problem-5-final.ipynb` and a fixed random seed for reproducibility.

3.2.1 Augmentation pipeline

The augmentation pipeline applies one randomly selected transformation per image. The implemented transforms (and parameter ranges) are:

- Rotate right: clockwise 5–30° (fill value 0)
- Rotate left: counter-clockwise 5–30° (fill value 0)
- Micro-rotate: small tilt ± 3–$5°$
- Translate: random shift $\Delta x, \Delta y$ up to ± 4 pixels
- Rotate + Translate: 5–20° plus small shift up to ± 3 pixels
- Gaussian white noise: σ in [0.02, 0.06] (image intensity scale)
- Salt-and-pepper: 2–5% pixel flips to 0 or 1
- Rotate then noise: combination of rotation and Gaussian noise

These are implemented in the notebook's `AUGMENTATION_FUNCS` set and are applied via a reproducible NumPy RNG per experiment.

3.2.2 LeNet-5 model (as used)

We use a compact LeNet-5 variant defined in the notebook. Layer summary (shapes assume input $1 \times 28 \times 28$):

- Conv1: Conv2d(1, 6, kernel=5, padding=2) – activation: Tanh; AvgPool2d(2)
- Conv2: Conv2d(6, 16, kernel=5) – activation: Tanh; AvgPool2d(2)
- Classifier: Flatten → Linear(16×5×5, 120) – Tanh → Linear(120,84) – Tanh → Linear(84,10)
- Loss: CrossEntropy; Optimizer: Adam; LR schedule: CosineAnnealingLR

3.2.3 Evaluation metrics

We report two standard metrics for multiclass classification plus wall-clock training time.

- **Accuracy**: fraction of correctly predicted examples,

$$\text{Accuracy} = \frac{1}{N} \sum_{i=1}^{N} \mathbf{1}\{\hat{y}_i = y_i\} \,,$$

where $\hat{y}_i$ is the predicted class, y_i the ground-truth, and N the number of examples.

- **F1 (macro)**: per-class F1 averaged across classes. Let for class c precision P_c and recall R_c be defined as usual; the per-class F1 is

$$F_{1,c} = \frac{2P_cR_c}{P_c + R_c},$$

and the macro score is

$$F1_{\text{macro}} = \frac{1}{C}\sum_{c=1}^{C} F_{1,c},$$

where C is the number of classes (10 here). Macro F1 weights all classes equally regardless of support.

3.2.4 Results

Below are the three result tables produced from the notebook outputs: Accuracy (%), F1 (macro) and Time (s). Replace the placeholder cells with the numeric outputs from the notebook `pivot_acc`, `pivot_f1`, and the timing values saved for each experiment.

Table 5: Accuracy (%) across augmentation levels and base dataset sizes

n_aug / n_real	**350 Real**	**750 Real**	**1000 Real**
0	82.5	92.0	91.5
1000	87.5	92.0	92.5
1500	88.0	93.0	93.0
2000	88.5	94.0	94.5

Table 6: F1 (macro) across augmentation levels and base dataset sizes

n_aug / n_real	**350 Real**	**750 Real**	**1000 Real**
0	0.8209	0.9202	0.9142
1000	0.8740	0.9199	0.9242
1500	0.8790	0.9310	0.9296
2000	0.8846	0.9395	0.9446

Table 7: Training time (s) across augmentation levels and base dataset sizes

n_aug / n_real	**350 Real**	**750 Real**	**1000 Real**
0	1.5	0.5	0.6
1000	1.3	1.4	1.6
1500	1.7	1.9	2.1
2000	2.2	2.4	2.6

3.2.5 Discussion

The empirical results in Tables 5 and 6 show consistent improvements from augmentation, with the largest relative gains at the smallest base size (350 examples). Quantitative observations:

- For 350 real examples accuracy increases from 82.5% (no augmentation) to 88.5% at 2000 augmented images (absolute gain 6.0%). F1 (macro) increases from 0.8209 to 0.8846 (gain 0.0637).

- For 750 real examples accuracy increases from 92.0% to 94.0% (gain 2.0%); F1 rises from 0.9202 to 0.9395 (gain 0.0193).
- For 1000 real examples accuracy increases from 91.5% to 94.5% (gain 3.0%); F1 rises from 0.9142 to 0.9446 (gain 0.0304).

Training time (Table 7) grows with the amount of augmentation. Example increases from no augmentation to 2000 augmented images:

- 350 real: 1.5 s -> 2.2 s (+0.7 s)
- 750 real: 0.5 s -> 2.4 s (+1.9 s)
- 1000 real: 0.6 s -> 2.6 s (+2.0 s)

Taken together, augmentation provides clear accuracy/F1 benefits, especially when the base dataset is small. However, marginal gains shrink as the real data pool grows. The 1500-2000 augmented range yields only small improvements over 1000–1500 in several cases, suggesting diminishing returns beyond 1500 synthetic images for this model/dataset combination.

3.2.6 Conclusions

The results indicate that controlled augmentation reliably improves classification performance for the LeNet-5 model on ReducedMNIST. Key takeaways:

- Augmentation yields the largest absolute benefit at the smallest base size (350): +6.0% accuracy and +0.0637 F1 when moving to 2000 augmented images.
- For medium/large base sizes (750, 1000) the gains are smaller (2–3% accuracy), implying that augmentation is most valuable when real data is scarce.
- Training time increases with augmentation; the 1500-2000 range yields limited extra performance versus the 1000–1500 range, so 1500 is a reasonable compromise for this setup.

Overall recommendation: apply augmentation (1000–1500 synthetic images) when limited to a few hundred real samples, and re-evaluate marginal gains before scaling beyond 1500 synthetic samples per experiment.

3.3 Problem 6: Generative Adversarial Networks (GAN)

Part 1 — Overview

We trained a conditional GAN on a reduced MNIST-style dataset (350 training examples, stratified to 35 examples per class) and used the trained generator to produce three samples per digit. Below we describe the network architecture and hyperparameters, report the generated samples, comment on timing and compute, and discuss training dynamics and limitations.

Architecture and design choices

Generator The generator is a conditional model that receives a noise vector $z \in \mathbb{R}^{100}$ concatenated with a learned label embedding (embedding dimension 50) and produces a single-channel 28×28 image using a small DCGAN-style decoder. Key layers and roles:

- **Embedding:** maps discrete label IDs to a continuous vector so the generator can condition image synthesis on class.
- **Linear projection + Unflatten:** projects the combined latent+label vector into a 3D feature map (512×4×4) that is amenable to transposed convolution upsampling.
- **ConvTranspose2d blocks:** successive transposed convolutions (512→256→128→64) with BatchNorm and ReLU upscale spatial resolution while stabilizing training and enabling richer representations.
- **Final Conv2d + Tanh:** crops/resamples to the target 28×28 size and outputs pixel values in $[-1, 1]$ for compatibility with the tanh output scaling.

Discriminator The discriminator is a conditional CNN receiving two channels: the image and a label "map" created by projecting the label embedding to a $1 \times 28 \times 28$ plane and concatenating it with the image. Notable design points:

- **Label projection:** lets the discriminator evaluate image+class consistency by providing class information spatially aligned with the input image.
- **Spectral normalization:** applied to convolution and linear layers to stabilise the critic and keep the Lipschitz constant bounded, which helps training stability when using high-capacity networks on small datasets.
- **LeakyReLU + BatchNorm:** standard DCGAN-style blocks to increase representational power while maintaining gradient flow.

Why these choices? The conditional setup (embedding + label map) improves per-class control over synthesis, which is essential when training from only 350 images. The generator architecture mirrors DCGAN upsampling best practices and keeps computation small. Spectral normalization in the discriminator reduces mode collapse and stabilizes adversarial training on a small dataset.

Hyperparameters

Following values were used (see the code for exact settings):

- Latent dimension: 100
- Label embedding: 50
- Epochs: 200

- Batch size: 32
- Optimizer: Adam, LR $= 2 \times 10^{-4}$, $\beta = (0.5, 0.999)$
- Loss: binary cross-entropy on logits (BCEWithLogitsLoss)
- Initialization: DCGAN-style normal init
- Regularization: spectral normalization on D

Training dynamics and observations

- **Discriminator vs Generator balance:** the discriminator is trained one step per generator step. Early in training the discriminator loss drops quickly while the generator loss rises; as training proceeds losses oscillate but the visual quality improves when the generator learns class-conditional features (strokes, thickness, rough digit shapes).
- **Mode collapse and diversity:** with only 35 examples per class, the model is vulnerable to limited diversity — spectral norm helps, but some digits still show lower diversity and tend to reproduce a few common prototypes.
- **Overfitting risk:** small dataset and high-capacity networks can make the discriminator memorize real examples; monitoring visual samples and simple pixel statistics can help detect this.
- **Practical checks:** we saved per-digit pixel mean/std (reported by the generation script) and visually inspected a 10×3 grid (10 digits × 3 samples) as the primary qualitative check.

Generated samples

Figure 5 shows the generator output (3 samples per digit) obtained after training on the 350-example reduced dataset. The image was produced by the notebook and saved as `gan_generated_samples.png`.

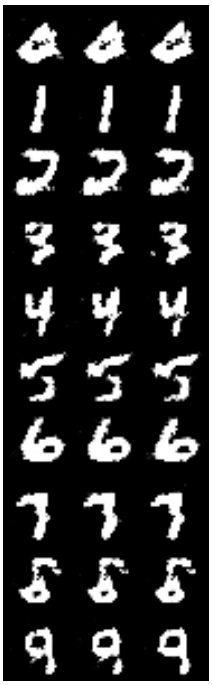


Figure 5: Generated samples from the conditional GAN: 10 rows (digits 0–9) $\times$ 3 columns (independent samples).

Visual assessment Most digits capture the coarse stroke patterns and relative topology of the target class (loops for 8, open curves for 2, vertical strokes for 1). However, finer details (consistent stroke width, crisp corners) are sometimes blurred and a few digits show mode-reduced variants (similar shapes repeated). This is expected given the limited real-data pool.

Timing and compute environment

The notebook records timestamps for major stages (dataset load, training, and generation) using a lightweight `Timer` context manager. Measured durations from the notebook run were:

- **Loading dataset:** 32.2 s
- **Training (200 epochs):** 48.2 s (about 0.80 min)
- **Generation (30 images + grid save):** 0.1 s

Computer setup The notebook output for `print_system_info()` was:

- OS: Linux 6.6.113+
- Python: 3.12.12
- PyTorch: 2.10.0+cu128
- CPU: Intel(R) Xeon(R) CPU @ 2.00GHz
- RAM: 31Gi total, 1.2Gi used
- GPU 0: Tesla T4, 14.6 GB VRAM
- GPU 1: Tesla T4, 14.6 GB VRAM
- CUDA: 12.8

The setup used for this run was a CUDA-enabled environment with two Tesla T4 GPUs on a Linux host, which explains the short training time despite using a relatively large generator.

Evaluation and limitations

Quantitative evaluation on such small datasets is challenging. Suggested additions for a more complete study:

- Compute FID/Inception-style metrics adapted for grayscale digits (requires a small reference set or pretrained feature extractor).
- Use a held-out classifier accuracy test: augment training with generated images and measure downstream classifier performance.
- Track diversity metrics (e.g., per-class nearest-neighbour distances in latent/image space).
- Save model checkpoints and generator seeds to reproduce and compare samples deterministically.

4 Appendices

.1 Code Snippets